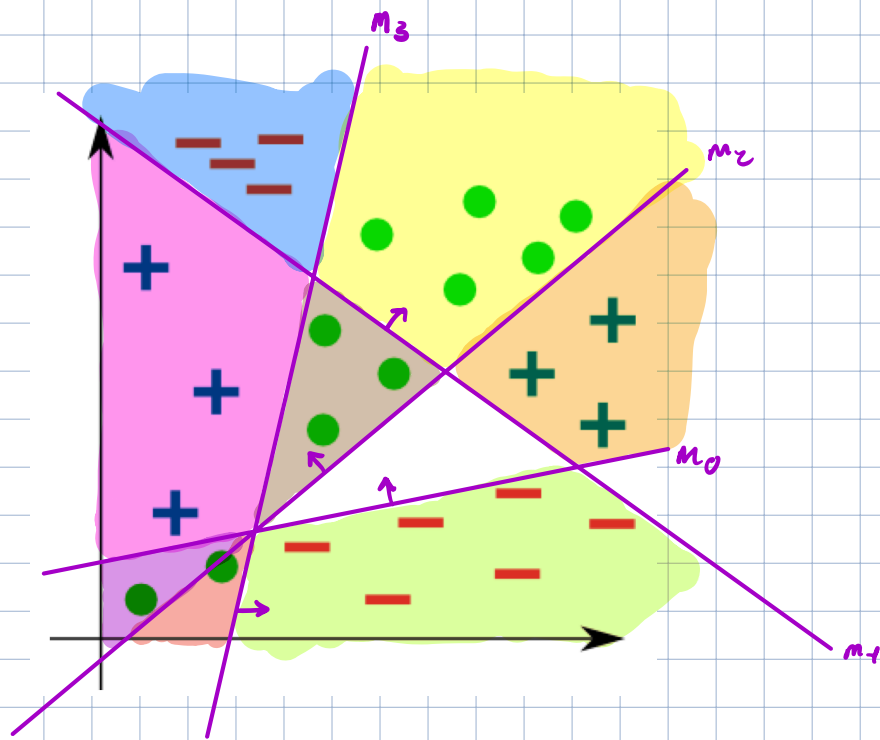




$M=4$

$$\begin{aligned} c^1 &:= \text{green circle} = (0, 0, 0, 1) & c^8 &:= \text{blue circle} = (1, 0, 0, 1) \\ c^2 &:= \text{pink circle} = (0, 0, 1, 0) & c^6 &:= \text{yellow circle} = (1, 1, 1, 1) \\ c^3 &:= \text{light purple circle} = (0, 0, 0, 0) & c^7 &:= \text{light orange circle} = (1, 0, 1, 0) \\ c^4 &:= \text{light green circle} = (1, 1, 1, 0) & c^5 &:= \text{light orange circle} = (1, 1, 0, 1) \end{aligned}$$

Layer 1 creates decision boundaries using a $\psi(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$

We receive decision boundary regions visualised above.

Layer 1 therefore produces a 4 dimensional vector Z_1 which layer 2 consumes after. Given only 8 out of the 9 regions are required for a perfect match, layer 2 only contains 8 perfect match neurons, one for every feature containing corner:

$$\forall i \in 1 \dots 8: \quad Z_1 \rightarrow h(Z_1; c_i) := \psi(z \cdot \beta_{MA}^i + b_{MA}^i) = \begin{cases} 1, & z = c \\ 0, & \text{otherwise} \end{cases}$$

$$\beta_{MA}^i = 2c^i - 1$$

$$b_{MA}^i = 0.5 - \sum_{j=1}^4 c_j^{(i)}$$

This means our layer 2 outputs a 8 dimensional vector Z_2

where only one item is 1.

Layer 3 consumes z_2 and feeds them into a set of masked logical ORs.

For the class green circles: •

Mask:

$$m^{\bullet} = (0, 1, 1, 0, 1, 1, 0, 0)^T$$

$$\beta_{\text{MOR}}^{\bullet} = m^{\bullet}$$

$$b_{\text{MOR}}^{\bullet} = -1/2$$

$$z_3^{\bullet} = \Psi(z_2 \cdot \beta_{\text{MOR}}^{\bullet} + b_{\text{MOR}}^{\bullet})$$

For +

$$m^{+} = (0, 0, 0, 0, 0, 1, 1, 1)$$

$$\beta_{\text{MOR}}^{+} = m^{+}$$

$$b_{\text{MOR}}^{+} = -1/2$$

$$z_3^{+} = \Psi(z_2 \cdot \beta_{\text{MOR}}^{+} + b_{\text{MOR}}^{+})$$

For

$$m^{-} = (1, 0, 0, 1, 0, 0, 0, 0)$$

$$\beta_{\text{MOR}}^{-} = m^{-}$$

$$b_{\text{MOR}}^{-} = -1/2$$

$$z_3^{-} = \Psi(z_2 \cdot \beta_{\text{MOR}}^{-} + b_{\text{MOR}}^{-})$$

Proof: Induction over L

LA: $L=2$

Since $\phi_1 = \text{id}$ one gets with $z_0 = X$

$$\begin{aligned} z_2 &= \phi_2(z_1 \cdot B_2 + b_2) \\ &= \phi_2(\phi_1(z_0 \cdot B_1 + b_1) \cdot B_2 + b_2) \end{aligned}$$

$$= (X \cdot B_1 + b_1) \cdot B_2 + b_2$$

$$= X \underbrace{B_1 B_2}_{:= B} + \underbrace{b_1 B_2 + b_2}_{:= b}$$

$$= XB + b$$

$\Rightarrow$ The 2-layer network is equivalent to a 1-layer network with $B = B_1 B_2$,

$$b = b_1 B_2 + b_2$$

II: Suppose it holds for ~~any~~ some fixed $L \geq 2$

IS: $L \rightarrow L+1$

$$z_L = \phi_L(z_{L-1} \cdot B_L + b_L)$$

$$= z_{L-1} B_L + b_L$$

$$\stackrel{\text{IH}}{=} (XB + b) B_L + b_L$$

$$= X \underbrace{B B_L}_{:= \tilde{B}} + \underbrace{b B_L + b_L}_{:= \tilde{b}}$$

$$= X \tilde{B} + \tilde{b}$$

$\Rightarrow$ The L -layer network is equivalent to a 1-layer network with matrix $\tilde{B}$ and bias $\tilde{b}$